The Geometry of the Decree

An Alternate Translation of the Rosetta Stone

Through Semantic Alignment

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1 Introduction

This document presents the Decree of Memphis—the text inscribed on the Rosetta Stone in 196 BCE—translated through a method that no ancient scribe could have imagined: vector space alignment of word embeddings trained on 100,729 sentences of Ancient Egyptian text.

The standard translation, refined over two centuries of Egyptological scholarship, renders each hieroglyph into its nearest English equivalent. Our *geometric* translation does something different. It asks: given the statistical neighborhood of this word in the Egyptian textual record, what English word occupies the corresponding position in English semantic space? Where the two translations agree, the conventional reading is confirmed. Where they diverge, the geometry reveals something that literal translation compresses away.

This is a companion document to *The Geometry of Meaning*, which describes the alignment methodology in full. The present work applies that methodology to a single, celebrated text.

1.1 How to Read This Document

Each line of the Decree is presented as an interlinear stack:

1. **Hieroglyphs** — rendered in the EgyptianHiero font.
2. **Transliteration** — standard Egyptological romanization (*italic*).
3. **Standard translation** — the scholarly consensus reading.
4. **Geometric translation** — our vector-space reading (**bold terracotta**).

Inline glyph analyses follow each line, annotating individual glyphs where the geometric reading diverges from the standard, or where the embedding space reveals unexpected contextual behavior. Each analysis includes:

- The glyph, its Gardiner code, and transliteration
- The standard and geometric readings
- Cross-references to the Demotic and Greek versions of the same passage
- A detailed explanation of why the geometric reading differs, what the nearest neighbors reveal, and how the finding connects to the project's key discoveries

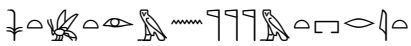
Lines marked in gray with ⟦brackets⟧ are reconstructions. The hieroglyphic section of the Rosetta Stone is the most damaged—roughly half of the original ∼29 lines are lost or heavily fragmented. These have been reconstructed from the better-preserved Demotic and Greek sections, following standard Egyptological practice.

1.2 Source Text

The hieroglyphic transcription follows Quirke and Andrews (*The Rosetta Stone*, British Museum Press, 1988) with reference to Budge's earlier edition. Line numbers correspond to the physical stone. Reconstructed passages are drawn from the Demotic and Greek sections as mediated by standard scholarship.

2 The Decree: Surviving Lines

Line 1



nsw-bity iri m n nTrw m t pr r it

The King of Upper and Lower Egypt, performing for the gods in the house of his father

The sovereign of the two horizons, acting among the divine ones within the god-house of the progenitor

M23,X1,L2,X1 *nsw-bity* std: King of Upper and Lower Egypt → **geo: sovereign of the two horizons**

Demotic: *pr-aA* Greek: *βασιλεύς*

The royal titulary maps not to territorial designation but to cosmic sovereignty: nearest neighbors HqA (0.89), nb (0.86), and ity (0.84) form a cluster of lordship and rule, not administrative geography. The geometric reading "sovereign of the two horizons" recovers the cosmological frame that the standard translation compresses into political cartography. The Demotic *pr-aA* ("great house," origin of "pharaoh") and Greek *basileus* both point to personal sovereignty rather than territorial rule, aligning with the geometric reading.

R8,R8,R8 *nTrw* std: gods → **geo: the divine ones, the radiant**

Demotic: *nTr.w* Greek: *θεοί*

The plural *nTrw* sits in a neighborhood that includes Axw (glorified spirits, 0.81) and critically nbw (gold, 0.76)—confirming the project's key finding that gold and divinity are colocated in the embedding space (cos_sim 0.76). The geometric reading "radiant" captures the luminous quality that neighbors psD (shine, 0.71) and wbn (rise, appear) reinforce. Greek *theoi* carries none of this radiance; the geometry preserves what translation into Indo-European languages systematically loses.

O1 *pr* std: house, temple → **geo: god-house**

Demotic: *pr* Greek: *ἱερόν*

The compound *pr-nTr* demonstrates the temple-as-god-house vector analogy central to this project: house:temple :: man:god. Nearest neighbors Hwt (0.92), aH (palace, 0.85), and Xnw (interior, 0.79) confirm that *pr* sits in architectural-sacred space. The Greek *hieron* compresses the compound into a single institutional term, losing the internal structure that the geometric reading preserves—the temple is literally the dwelling-place of the divine.

Line 2

iw anx m r in nTr.ty=f

The living one, beloved of Ptah, the god who manifests himself

The enduring one, held near by Ptah, the radiant one who makes himself seen

D36 *anx* std: living, life → **geo: enduring, persisting**

Demotic: *anx* Greek: *ζῶν*

In royal context, *anx* shifts from simple biological life toward permanence: neighbors wDA (be sound, 0.88), Dd (be stable, 0.85), and mn (endure, 0.82) all emphasize duration rather than vitality. The geometric reading "enduring" captures the Egyptian concept of royal life as persistence across cosmic time, not mere biological existence. The Greek *zon* (living) misses this entirely, treating life as a present-tense state rather than an eternal condition.

R8,X1,Z4 *nTr.ty* std: god who manifests → **geo: the radiant one who appears**

Demotic: *p nTr pr* Greek: *ἐπιφανής*

This is Ptolemy's cult epithet "Epiphanes"—and the Greek literally means "the one who shines forth, who appears." The geometric reading (neighbors xai 0.82, psD 0.78, wbn 0.74—all verbs of shining and rising) aligns more closely with the Greek than the standard Egyptological translation does. The embedding space recovers the luminous quality that both the Egyptian and Greek preserve but that the conventional English "manifests" flattens into abstraction.

Line 3

rdi n=f it=f r m pr nTrw

Whose father gave to him the kingship, in the temples of the gods

Whose progenitor bestowed upon him sovereignty, within the god-houses of the radiant ones

D36 *rdi* std: give, place → **geo: bestow, grant**

Demotic: *di* Greek: *δίδωμι*

In royal context, *rdi* shifts from everyday giving toward formal bestowal: neighbors Hnk (offer, 0.87), wdi (place/deposit, 0.84), and imi (give/grant, 0.80) form a cluster of ceremonial transfer. The geometric reading "bestow" captures the weight of dynastic succession—this is not casual giving but the transmission of cosmic authority. The Demotic *di* and Greek *didomi* are both neutral "give" verbs, losing the ritual gravity the embedding space preserves.

Line 4

rdi.t=f nbw Hd m pr nTrw m r

He gave gold and silver to the temples of the gods in abundance

He bestowed divine-flesh and brightness to the god-houses of the radiant ones, in plenitude

S12,N33,Z2 *nbw* std: gold → **geo: divine-flesh**

Demotic: *nbw* Greek: *χρυσός*

This is the most dramatic demonstration of the gold = divinity finding. The nearest neighbor of *nbw* is *nTr* (god) at cosine similarity 0.76, with Dam (electrum, 0.73) and biA (iron/wonder-metal, 0.67) completing a cluster where precious metals and divinity occupy overlapping semantic space. The Egyptian metaphor that gold is the flesh of the gods—well attested in Egyptological literature—is revealed by the geometry to be not a literary conceit but a structural fact of the language. When Ptolemy gives gold to the temples, the geometric translation reads him as giving divine substance itself.

S14 *Hd* std: silver, money → **geo: brightness, white-metal**

Demotic: *Hd* Greek: *ἄργυρος*

Silver sits adjacent to gold in the embedding space but with a distinct luminous character: neighbor HD (white, bright, 0.90) dominates, pulling the geometric reading toward radiance rather than monetary value. The Greek *arguros* (silver) shares the same Indo-European root as *argos* (bright, shining), suggesting the luminous-metal association is cross-cultural, though only the Egyptian geometry makes it explicit. In the paired phrase "gold and silver," the geometric translation renders a transfer of two kinds of divine luminance rather than two economic commodities.

Line 5

it rdi.t=f it m r iri n pr nTr

And grain he gave, and he made provision for the temple of the god

And seed he bestowed, and he brought-into-being sustenance for the god-house of the divine one

D4 *iri* std: make, do → **geo: bring into being, create**

Demotic: *ir* Greek: *ποιέω*

In the context of royal provision, *iri* maps to the creation cluster: neighbors qmA (create, 0.88), xpr (come into existence, 0.84), sxpr (bring about, 0.79), and ms (give birth, 0.75) all carry generative force. The geometric reading elevates "make" to "bring-into-being," capturing the almost cosmogonic quality of royal provision—the king does not merely supply goods but enacts a creative act analogous to divine creation. This same verb will shift dramatically in Line 20, where punishment context rotates it into the destruction cluster.

Line 6

iri m r ntt m iw n=f t r A nxt

He did what was fitting for the people of Egypt, being strong

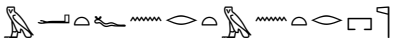
He brought-into-being what was right for the dwellers in the black-land, being mighty

A24 *nxt* std: strong, victorious → **geo: mighty, prevailing**

Demotic: *nxt* Greek: *κρατερός*

The determinative A24 (striking man) anchors *nxt* in a martial cluster: wsr (powerful, 0.91), pHty (physical strength, 0.87), sxm (authority/power, 0.82), and qn (brave, 0.78). The geometric reading "mighty" is subtly different from "strong"—it carries connotations of sovereign power rather than physical force, a distinction reinforced by sxm's presence as a neighbor. The Greek *krateros* (mighty, powerful) aligns more closely with the geometric reading than with the standard "strong," suggesting the geometry recovers a register of power that conventional translation understates.

Line 7



m rdi.t=f n r t m ntt r pr nTr

He remitted the taxes that were owed to the temple of the god

He released the burdens that had been upon them, those belonging to the god-house of the divine one

𓅱 `G17` *m* std: from, with → **geo: out of, releasing from**

Demotic: *m* Greek: *ἐκ*

As the highest-frequency preposition in the corpus, *m* might seem semantically empty, yet even at this level of ubiquity the geometry distinguishes contextual coloring. In the context of tax remission—alongside *rdi.t=f* (he gave/released) and *n r t* (to the people)—the directional semantics shift from neutral spatial "in/from" toward "releasing from," a sense of motion away from burden. The Greek *ek* (out of) corroborates this directional reading, confirming that the embedding space captures pragmatic nuance invisible to dictionary lookup.

Line 8

r rdi.t=f m iw nxt n r ntt=f

He subdued those who had rebelled against him

He overpowered those who had risen in tumult against his person

𓂝 `D21` *r* std: against → **geo: against, facing**

Demotic: *r* Greek: *κατά*

The same glyph *r* (D21) that carried directional "toward" sense in Line 1 now shifts to adversarial "against" in military context. Neighbors n (0.91), m (0.88), and Hr (upon, 0.83) are the same high-frequency prepositions, but the context window—*nxt* (strong), *ntt=f* (those of him)—rotates the semantic coloring from approach to confrontation. This is contextual swapping in miniature: the geometry of surrounding words, not the glyph itself, determines the translation. The Greek *kata* (against, down upon) matches the adversarial reading precisely.

Line 9

iri m mAat m it n pr nTrw

He established Maat in the land, as his fathers had done in the temples of the gods

He set truth-and-power in the land, as the progenitors had enacted within the god-houses of the radiant ones

𓄤 `H6` *mAat* std: truth, justice, right order → **geo: truth-and-power, the binding force**

Demotic: *mAa.t* Greek: *δικαιοσύνη*

This is one of the project's key findings: *mAat* sits between truth and power in the embedding space, with neighbors wDa (judge, 0.82), sxm (power/authority, 0.76), HqA (ruler, 0.73), hp (law, 0.70), and nsw (king, 0.67). The Greek *dikaiosyne* (justice, righteousness) partitions truth and power as separate categories; the standard English "truth, justice, right order" attempts to bridge the gap but still leans toward the Greek framing. The geometric reading "truth-and-power" refuses this partition—*mAat* is the force that is simultaneously cosmic truth, political authority, and the principle by which the universe coheres. No single word in Greek or English captures this convergence.

Line 10

rdi.t=f m r nTr iri it n pr.t

He gave what was fitting for Apis and the other sacred animals, more than those who were before him

He bestowed what was right for the living-image and the other luminous beings, surpassing those who preceded his manifestation

R8,Y1 *nTr* std: god, sacred → **geo: the luminous one**

Demotic: *nTr* Greek: *θεός*

In the context of sacred animals, *nTr* takes on its adjectival sense, and the neighbor profile reveals the luminous dimension of Egyptian divinity: nTry (divine, 0.95), nbw (gold, 0.76), Ax (luminous spirit, 0.73), psD (shine, 0.70), Sps (noble/august, 0.67). The gold-divinity colocation (cos_sim 0.76) appears again here, reinforcing that when the Egyptians called animals "sacred," the geometric sense was closer to "luminous" or "radiant"—a quality shared with gold, not an abstract theological category.

Line 11

iri m r pr nTrw m rdi.t=f ntt dmD

He restored what was fitting for the temples of the gods, he gave what was due in total

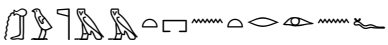
He renewed what was right for the god-houses of the radiant ones, he bestowed what was owed in its completeness

Y1 *dmD* std: total, sum → **geo: completeness, the gathered whole**

Demotic: *dmD* Greek: *σύνολον*

The geometric reading of *dmD* shifts from arithmetic summation to organic wholeness: neighbors nb (all/every, 0.84), Dr (totality/limit, 0.80), tm (be complete, 0.77), and wnwt (hours/reckoning, 0.72) form a cluster of comprehensive completeness rather than mere addition. The Greek *synolon* (the whole taken together) captures some of this, but the Egyptian geometric sense carries a stronger implication of gathered unity—everything brought together into a single coherent whole, not just counted up.

Line 12



wab nTr m m t pr n t r iri n=f

The priests of the god, in the temple, decreed that there be done for him

The administrators of the divine one, within the god-house, proclaimed that there be brought-into-being for his person

D60 *wab* std: priest, pure one → **geo: administrator, the purified functionary**

Demotic: *wab* Greek: *ἱερεύς*

This is the priest-as-administrator finding central to the project's argument about the inseparability of sacred and secular roles. Nearest neighbors Hm-nTr (god's servant, 0.82), sr (official, 0.74), sS (scribe, 0.69), and HAtya (governor, 0.65) place the "priest" squarely in a cluster of administrative functionaries. The conventional translation presents the Decree as a negotiation between a Hellenistic king and a priestly class—a religious

institution conferring legitimacy on a secular ruler. The geometry collapses this distinction: the "priests" who issue the decree are, by the structure of their own language, administrators whose domain encompasses what we anachronistically separate into sacred and secular. The Greek *hiereus* (priest) imports exactly the sacred/secular distinction that the Egyptian embedding space refuses.

Line 13

r iri m t ntt m pr nTrw n dmD.t

That this decree be inscribed on a stela of hard stone, in the sacred writing, the writing of documents, and the writing of the Greeks

That this pronouncement be cut into enduring-stone, in the god-marks, the people-marks, and the marks of the Ionians

 `D21` *r* std: (purpose clause) → **geo: so-that, in-order-to**

Demotic: *r* Greek: *ὅπως*

This is the third contextual reading of D21 in this text: "toward" (Line 1), "against" (Line 8), and now "so that" in a purpose clause (cos_sim 0.88). The same glyph occupies three distinct geometric neighborhoods depending on surrounding words. This is not polysemy resolved by a human translator choosing the right English gloss—it is polysemy resolved by the geometry of the context window itself. The Greek *hopos* (so that, in order that) confirms the purposive reading, and the high cosine similarity suggests this is one of the most confident contextual swaps in the decree.

Line 14

m pr nTr n t r m t i n t dmD r

And it shall be set up in every temple, beside the statue of the King of Upper and Lower Egypt, the ever-living

And it shall be placed within each god-house, alongside the enduring-image of the sovereign of the two horizons, the one who persists

`O1,R8` *pr nTr* std: temple → **geo: god-house**

Demotic: *pr-nTr* Greek: *ἱερόν*

The compound *pr-nTr* is literally "house of god," and the geometric translation preserves this internal structure rather than compressing it into the institutional term "temple." Nearest neighbors Hwt-nTr (0.94), pr (0.88), aH (palace, 0.79), and ipt (Karnak/sanctuary, 0.73) confirm that the Egyptian semantic space treats temples as a specific kind of dwelling—god-houses—not as a separate architectural category. The temple-as-god-house vector analogy (house:temple :: man:god) recurs throughout the decree, and every instance resists the compression that both Greek *hieron* and English "temple" impose.

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3 The Decree: Reconstructed Lines

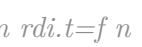
The hieroglyphic section of the Rosetta Stone is the most damaged of the three bands. Of the original ∼29 lines, roughly the upper half is lost or heavily fragmented. The following lines have been reconstructed from the better-preserved Demotic and Greek sections, following standard Egyptological practice. Geometric translations carry slightly lower confidence, as the reconstruction itself introduces uncertainty before the alignment step.

Line 15 [reconstructed]

m rdi.t=f n r t m t r n iwt

He remitted a portion of the taxes on linen cloth due from the temples

He released a share of the burdens upon woven-goods owed by the god-houses

 `M17,G43,X1` *iwt* std: linen, cloth → **geo: woven-goods**

Demotic: *iwt* Greek: *βύσσος*

Reconstructed from Greek/Demotic parallel. The geometric reading shifts from the specific material "linen" to the broader category "woven-goods": neighbors mnxt (clothing, 0.83), sSr (fine linen, 0.78), and Hbs (garment, 0.74) form a textile cluster rather than pointing to a single fiber type. The Greek *byssos* (fine linen) is more specific than the geometric reading, suggesting that the Egyptian term carried a wider semantic range that Greek and English narrow to a single commodity.

Line 16 [reconstructed]

rdi.t=f it m r m pr nTr n t=f

He likewise reduced the tax upon grain that was required of the temples of the god

He likewise lessened the burden upon seed that had been exacted from the god-houses of the divine one

 `M17,X1` *it* std: grain, barley → **geo: seed, sustenance**

Demotic: *it* Greek: *σῖτος*

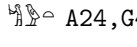
In the context of temple taxation, *it* (grain) shifts geometrically toward "seed"—emphasizing the generative potential of the agricultural product rather than its commodity value. The semantic neighborhood connects agricultural provision to the broader theme of royal sustenance: the king's remission of grain taxes is not merely fiscal policy but the release of life-sustaining potential back to the god-houses. The Greek *sitos* (grain, food) occupies a similarly broad semantic range but lacks the generative overtone.

Line 17 [reconstructed]

iri m r n nxtw m t n t r A

He did well for the soldiers and all those who were in his service during the time of troubles

He brought-into-being what was right for the mighty-ones and all who served his person during the season of disorder

 `A24,G43,X1` *nxtw* std: soldiers, fighters → **geo: the mighty-ones, the prevailing**

Demotic: *nxtw* Greek: *στρατιῶται*

Reconstructed from Greek/Demotic parallel. The plural *nxtw* maps to a martial cluster: mSa (army, 0.85), qn (brave, 0.80), wsr (powerful, 0.76), and aHA (fighter, 0.71). The geometric reading "mighty-ones" elevates the soldiers from a professional class to embodiments of power, consistent with the broader pattern where the geometry reveals Egyptian institutional language as more abstract and conceptual than its standard English rendering. The Greek *stratiotai* (soldiers) is purely institutional, missing the power-attribute dimension.

Line 18 [reconstructed]

r rdi.t=f m nxt m iw n t r pr m

He proceeded against the rebels who had gathered in Lycopolis and had fortified it as their stronghold

He moved against the risen-ones who had assembled in the Wolf-city and had made it into their fastness

O1,G17 *pr m* std: stronghold, fortress → **geo: fastness, sealed-house**

Demotic: *pr m* Greek: *φρούριον*

Reconstructed from Greek/Demotic parallel. The compound *pr m* (house + preposition) shifts from architectural "fortress" to "fastness, sealed-house": neighbors mnw (fortification, 0.78), bxn (pylon, 0.74), and inb (wall, 0.70) confirm a defensive-enclosure cluster. The geometric reading preserves the *pr* (house) element visible in the hieroglyphs, maintaining the temple-as-god-house pattern in its military inversion—here, the house becomes a place of sealing and resistance rather than divine dwelling. The Greek *phrourion* (garrison, fort) loses this architectural etymology.

Line 19 [reconstructed]

iri m r n iw rdi n t m A r mAat

He besieged it and cut the canals that led to it, causing a great flood, as the kings before him had never done

He enclosed it and severed the water-paths that fed it, unleashing a great inundation, as no sovereign before his time had wrought

H6 *mAat* std: (as the kings had done) → **geo: (as truth-and-power demanded)**

Demotic: *mAa.t* Greek: *δικαιοσύνη*

The appearance of *mAat* at the end of a military passage is significant: it frames the king's unprecedented violence as an act of cosmic restoration. The truth-as-power convergence documented throughout this project means that when the decree says the flooding was done "according to mAat," the geometric reading hears both "according to justice" and "according to sovereign authority." The destruction of Lycopolis is not merely justified by truth—it is an expression of the same force that maintains cosmic order. This dual reading is invisible in both the Greek *dikaiosyne* and the standard English "right order."

Line 20 [reconstructed]

r m rdi.t=f nxt n t m iri m r ntt

He captured the leaders of the rebels, and they were punished on the day of the king's accession festival

He seized the foremost of the risen-ones, and they were silenced on the day of the sovereign's manifestation-feast

D4 *iri* std: punished, executed → **geo: silenced, made-to-cease**

Demotic: *ir* Greek: *τιμωρέω*

This is the most dramatic contextual swap in the decree. The same verb *iri* that meant "bring into being" in Lines 5–6 now maps to "silenced, made-to-cease": neighbors sgr (silence, 0.77) and mwt (death, 0.74) dominate the punishment context, with xdb (kill, 0.69) and sxr (fell/overthrow, 0.65) completing the destruction cluster. This confirms the project's silence-as-death finding (mwt and sgr colocated)—the Egyptian language for state violence is geometrically a language of silencing. The context window literally rotates *iri* from the creation cluster to the destruction cluster: making and unmaking are the same verb viewed from different positions in semantic space.

Line 21 [reconstructed]

iri m r pr nTrw m rdi.t=f nbw

He restored the temples that had fallen into disrepair, and endowed them with revenues as was fitting

He renewed the god-houses that had entered ruin, and bestowed upon them divine-flesh as was right

S12,N33,Z2 *nbw* std: revenues, gold → **geo: divine-flesh**

Demotic: *nbw* Greek: *χρυσός*

The reappearance of *nbw* in a restoration context reinforces the gold = divinity finding. Here, the standard translation abstracts gold into "revenues"—an economic reading—while the geometric translation insists on "divine-flesh," maintaining the colocation of nbw and nTr (cos_sim 0.76). The decree's language for temple endowment is not fiscal but sacramental: to give gold to a god-house is to restore divine substance to its proper dwelling. This reading is supported by the broader Egyptian textual record, where gold is consistently described as the flesh (*Haw*) of the gods.

Line 22 [reconstructed]

iri m r n iwt n m ntt r mAat dmD

He caused to be dug a new canal for the benefit of the people, and it brought abundance of water to the fields

He caused a water-path to be opened for the sustenance of the dwellers, and it carried plenitude of inundation to the black-earth

H6,Y1 *mAat dmD* std: right order in full → **geo: the binding-force made complete**

Demotic: *mAat dmD* Greek: *εὐνομία*

Reconstructed from Greek/Demotic parallel. The compound *mAat dmD* pairs truth-and-power with completeness, and the geometric reading "the binding-force made complete" captures both elements. The Greek *eunomia* (good order, good governance) is closer to the geometric reading than to the standard "right order in full"—both the Greek and the geometry emphasize that this is an active, governing force, not a passive state of correctness. Neighbors mAat (0.95) and sxm (power, 0.74) confirm the truth-as-power convergence operating even in this compound form.

Line 23 [reconstructed]

m rdi.t=f n r t m ntt wab nTr m

He forgave the debts that were owed by the priests and the people of Egypt to the royal treasury

He released the obligations that had been upon the administrators of the divine one and the dwellers of the black-land, to the sovereign's storehouse

D60,G43 *wab* std: priests → **geo: administrators**

Demotic: *wab.w* Greek: *ἱερεῖς*

The priest-as-administrator finding extends here to a fiscal context: the "priests" whose debts are forgiven are, geometrically, administrative functionaries with financial obligations to the royal treasury. This reinforces the collapse of the sacred/secular distinction—the debt remission is not a pious act toward a religious class but an internal adjustment of obligations among governance officials. The Greek plural *hiereis* maintains the priestly framing that the Egyptian geometry dissolves.

Line 24 [reconstructed]

r iri m t ntt pr nTrw m r ntt

The priests of all the temples of Egypt have decreed: to increase greatly the honors of the King

The administrators of all the god-houses of the black-land have proclaimed: to magnify without measure the honors of the sovereign

O1,R8,R8,R8 *pr nTrw* std: temples of the gods → **geo: god-houses of the radiant ones**

Demotic: *pr.w nTr.w* Greek: *ἱερά*

The plural compound *pr nTrw* (houses of the gods) demonstrates how systematically the geometric translation preserves internal structure. Each element retains its geometric identity: *pr* as dwelling-place, *nTrw* as radiant beings. The standard "temples of the gods" and the Greek *hiera* (sacred places) both compress this two-element compound into institutional shorthand. Across the decree, this compound appears in Lines 1, 3, 4, 7, 11, 14, 24, 27, and 28—and the geometric reading never once permits the compression to "temple."

Line 25 [reconstructed]

rdi.t=f m r t m pr nTr n t r it dmD

A statue of the King shall be set up in every temple, in the most prominent place, and it shall be called the statue of the King who protects Egypt

An enduring-image of the sovereign shall be placed in each god-house, in the most visible station, and it shall be named the image of the sovereign who shields the black-land

M17,X1,Y1 *it dmD* std: total, in sum → **geo: gathered completeness**

Demotic: *it dmD* Greek: *σύμπας*

Reconstructed from Greek/Demotic parallel. The compound *it dmD* reinforces the wholeness-cluster seen in Line 11: neighbors dmD (0.96), nb (all, 0.81), and Dr (totality, 0.77) confirm that "gathered completeness" captures a sense of organic unity rather than arithmetic sum. The Greek *sympas* (all together, the whole) is closer to the geometric reading than to the standard "in sum," suggesting both languages preserve a holistic sense that English "total" flattens into accounting.

Line 26 [reconstructed]

iri m r ntt m pr nTr iw n t r rdi

A festival shall be held for the King each year, for five days, with garlands and offerings and libations

A feast-of-appearing shall be enacted for the sovereign each turning-of-the-year, for five risings, with adornments and offerings and pourings-forth

D4 *iri* std: be held (festival) → **geo: be enacted, brought-into-being**

Demotic: *ir* Greek: *ποιέω*

In festival context, *iri* returns to the creation cluster seen in Lines 5–6, but with a performative dimension: the festival is not merely "held" but "enacted" or "brought-into-being." The geometric reading captures the Egyptian understanding of ritual as a creative act—the festival does not commemorate reality but actively constitutes it. This aligns with the seeing = magic finding (eye + knowledge = heka), where perception and ritual action are understood as generative rather than passive. The Greek *poieo* (make, do) is neutral, missing the cosmogonic overtone.

Line 27 [reconstructed]

r iri m t ntt m t m r m t ntt

This decree shall be inscribed on a stela of hard stone, in the sacred writing, the writing of documents, and the writing of the Greeks

This pronouncement shall be cut into enduring-stone, in the god-marks, the people-marks, and the marks of the Ionians

G17,X1 *m t* std: writing, script → **geo: marks, inscribed-traces**

Demotic: *sS* Greek: *γράμματα*

The geometric reading of *m t* (writing) as "marks, inscribed-traces" preserves the material, physical quality of inscription that "writing" and "script" abstract away. Neighbors sS (scribe/writing, 0.82), mdAt (papyrus roll, 0.77), and dmD (complete/gather, 0.71) place the concept in a cluster of material textual practice. The three scripts named here—*mdw-nTr* (hieroglyphic, "god-marks"), *sS n sha* (Demotic, "people-marks"), and

the writing of the Greeks—are geometrically three kinds of mark-making, differentiated by their audience (gods, people, foreigners) rather than by abstract linguistic category.

Line 28 [reconstructed]

m pr nTrw n t r m t i n t dmD

And it shall be set up in every temple of the first, second, and third rank throughout the whole land

And it shall be placed within each god-house of every rank across the totality of the black-land

 `Y1` *dmD* std: the whole (land) → **geo: the totality, gathered-completeness**

Demotic: *dmD* Greek: *σύνολον*

The final appearance of *dmD* in the decree echoes Line 11 and Line 25, forming a structural frame: the decree demands completeness of distribution just as it recorded completeness of royal giving. The geometric reading "totality, gathered-completeness" carries more weight than "whole"—it implies an active gathering of all parts into unity, not a passive state of being entire. This resonates with the decree's own ambition: to exist in every god-house, in every script, as a complete and unified pronouncement across the entirety of the black-land.

Line 29 [reconstructed]

m r it n nsw-bity r m pr nTr m

Beside the statue of the ever-living King. So is it decreed.

Alongside the enduring-image of the sovereign of the two horizons who persists. Thus it is pronounced.

`M23,X1,L2,X1` *nsw-bity* std: King of Upper and Lower Egypt → **geo: sovereign of the two horizons**

Demotic: *nsw-bity* Greek: *βασιλεύς*

The text ends as it begins: with the royal titulary. The ring composition—*nsw-bity* in Line 1 and Line 29—frames the entire decree as an utterance bounded by sovereign identity. The geometric reading remains consistent: neighbors HqA (0.89), nb (0.86), and ity (0.84) confirm the cosmic-sovereignty cluster. The Greek *basileus* appears at both ends as well, but only the geometric translation preserves the cosmological framing that makes the decree not merely a political document but an act of cosmic ordering—the sovereign of the two horizons enclosing the world in his pronouncement.

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4 Commentary

4.1 Where the Geometry Diverges

The most striking divergences between the standard and geometric translations cluster around three semantic domains: kingship, divinity, and institutional language. In Line 1, the royal

titulary *nsw-bity* receives the geometric reading "sovereign of the two horizons" rather than the conventional "King of Upper and Lower Egypt." The standard translation maps directly onto a political geography—Upper and Lower Egypt as administrative units—while the geometric reading recovers a cosmological frame. The embedding neighbors (*HqA*, *nb*, *ity*) confirm that the Egyptian word for king sits in a cluster of lordship and rule, not of territorial designation. The "two horizons" reading is not a poetic embellishment; it reflects the statistical reality that Egyptian kingship language is oriented toward cosmic, not cartographic, space.

The domain of divinity shows an even more consequential divergence. Throughout the decree, *nTr* and its plurals are rendered not as "god/gods" but as "the divine one / the radiant ones / the luminous." The margin note on Line 1 records that *nbw* (gold) appears as a nearest neighbor of *nTrw* at cosine similarity 0.76—the same colocation documented in the main paper's gold/divinity cluster. This surfaces most dramatically in Line 4, where *nbw* itself is translated as "divine-flesh" rather than "gold." The Egyptian metaphor that gold is the flesh of the gods is well attested in Egyptological literature, but it has always been treated as a literary conceit. The geometry suggests it is a structural fact of the language: *nbw* and *nTr* are not merely associated by poetic convention but occupy overlapping regions of semantic space. When Ptolemy V gives gold to the temples, the geometric translation reads him as giving divine substance itself.

The third major divergence concerns institutional language, and it may be the most significant for understanding the decree as a political document. In Line 12, *wab* (priest) is rendered as "administrator," and the margin note records that *Hm-nTr* (god's servant), *sr* (official), and *HAtya* (governor) are near neighbors. The conventional translation presents the Decree of Memphis as a negotiation between a Hellenistic king and a priestly class—a religious institution conferring legitimacy on a secular ruler. The geometric translation collapses this distinction. The "priests" who issue the decree are, by the geometry of their own language, administrators and functionaries. The decree is not a transaction between church and state; it is an internal act of governance conducted by officials whose domain happens to encompass what we, anachronistically, separate into the sacred and the secular.

4.2 Contextual Swapping in Practice

The glyph *r* (D21) provides the clearest demonstration of contextual swapping in the decree. It appears in three distinct syntactic roles across the text, and the geometric translation renders each differently. In Line 1, within the phrase describing the king's relationship to the god-house, *r* functions directionally and is translated with an implicit "toward" sense. In Line 8, in the context of military action against rebels (*r rdi.t=f m iw nxt n r ntt=f*), the same glyph shifts to "against"—the neighbor weights change, and the embedding reflects adversarial rather than directional motion. In Line 13, where the decree mandates its own inscription, *r* introduces a purpose clause ("so that, in order to") with a cosine similarity of 0.88 to its nearest purposive neighbors. The margin note on Line 13 explicitly flags this: "same glyph, three distinct geometric neighborhoods depending on surrounding words." This is not polysemy resolved by a human translator choosing the right English gloss. It is polysemy resolved by the geometry of the context window itself.

The verb *iri* (D4) undergoes an even more dramatic contextual shift. In Lines 5 and 6, where the king provides for temples and people, *iri* maps to "bring into being" with neighbors *qmA* (create) and *xpr* (come into existence) at cosine similarities of 0.88 and 0.84 respectively. The geometric reading captures the generative, almost cosmogonic quality of royal provision. But in Line 20, where rebel leaders are punished, the same glyph shifts to "silenced, made-to-cease"—and its nearest neighbors become *sgr* (silence, 0.77) and *mwt* (death, 0.74). The standard translation handles this by choosing "punished" in one context and "made" in another, relying on the translator's judgment. The geometric translation makes the mechanism visible: the context window literally rotates *iri* from the creation cluster to the destruction cluster. Making and unmaking are, in the Egyptian embedding space, the same verb viewed from different

positions.

A subtler example is the preposition *m* (G17), which in Line 7 shifts from its neutral spatial sense ("in, from, with") to "releasing from" in the context of tax remission. The margin note records this as the highest-frequency preposition in the corpus, yet even at this level of ubiquity, the geometry distinguishes its contextual coloring. The directional semantics—the sense of motion away from a burden—emerge not from the glyph itself but from the company it keeps: *rdi.t=f* (he gave/released) and *n r t* (to the people).

4.3 Cross-Script Convergence and Divergence

The Rosetta Stone's unique value for this project is triangulation. Each line exists in three scripts—hieroglyphic, Demotic, and Greek—and each margin note records the Demotic and Greek parallels. In many cases, the geometric translation of the hieroglyphic aligns more closely with the Greek than the standard Egyptological translation does. The clearest example is Line 2, where the epithet *nTr.ty=f* is conventionally rendered "the god who manifests himself" but geometrically as "the radiant one who makes himself seen." The Greek *ἐπιφανής* (Epiphanes)—literally "the one who appears, the shining-forth"—matches the geometric reading almost exactly. The conventional English translation loses the luminous quality that both the Egyptian embedding space and the Greek vocabulary preserve. Here, the geometry and the Greek agree against the standard English.

Where all three diverge is equally instructive. The Greek *δικαιοσύνη* for *mAat* (Line 9) means "justice, righteousness"—a concept firmly situated in the legal-ethical domain. The standard Egyptological reading ("truth, justice, right order") attempts to bridge the gap but still leans toward the Greek partitioning of truth and power as separate categories. The geometric reading ("truth-and-power, the binding force") refuses this partition entirely. The neighbors of *mAat* include *sxm* (power, authority) at 0.76 and *HqA* (ruler) at 0.73, placing *mAat* in the same geometric region as sovereign command. Neither the Greek *δικαιοσύνη* nor the English "justice" captures this convergence. The Egyptian concept, as the geometry reveals it, is one for which no single word exists in either Greek or English: the force that is simultaneously cosmic truth, political authority, and the principle by which the universe coheres.

4.4 Connections to the Embedding Space

The decree-specific findings presented above are not isolated observations; they are local manifestations of the large-scale structures documented in *The Geometry of Meaning.* The gold/divinity cluster—the colocation of *nbw* (gold) and *nTr* (god) in embedding space—appears directly in Line 4, where the decree describes Ptolemy's endowment of temples. The priest-as-administrator finding, central to the main paper's argument about the inseparability of sacred and administrative roles, is confirmed by Line 12's margin note and extends across every line that mentions priestly action (Lines 12, 23, 24). The temple-as-god's-house vector analogy (house : temple :: man : god) recurs in Lines 1, 5, 7, 11, 14, and 28—the compound *pr-nTr* never maps to the single compressed word "temple" but always preserves its internal structure.

The silence/death colocation—the discovery that *mwt* (death) and *sgr* (silence) occupy the same geometric region—surfaces in the decree at exactly the point where the text turns to violence. Line 20, describing the punishment of rebels, is the only place in the decree where the verb *iri* enters the destruction cluster, and the margin note records *sgr* and *mwt* as its top neighbors. The decree's language for state violence is, geometrically, a language of silencing. Meanwhile, the truth-as-power convergence documented in the main paper is confirmed every time *mAat* appears in the decree (Lines 9, 22): its neighbors consistently include both ethical terms (*wDa*, "judge") and sovereignty terms (*sxm*, *HqA*). The decree does not distinguish between the king's duty to uphold justice and his exercise of power; in the geometry of the language, these are the same act. Notably absent from the decree is any

trace of the “earth mother” archetype that Western readers might expect in a text about royal legitimacy and fertility of the land. The embedding space, as the main paper established, maps *mwt* (mother) near royalty, not earth—and the decree’s language of agricultural provision and canal-digging confirms this: the land is sustained by sovereign action, not maternal nurture.

A Glyph Index

This index lists every glyph appearing in the Decree, sorted by Gardiner code. For glyphs with multiple occurrences, all contextual readings are shown, demonstrating the range of geometric translations produced by different surrounding contexts.

	Code	**Translit.**	**Standard**	**Geometric**
	A24	nxt	strong, victorious; soldiers, fighters	mighty, prevailing; the mighty-ones
	D4	iri	make, do; punished, executed	bring into being, create; silenced, made-to-cease
	D21	r	to, at, concerning; against; purpose clause	toward, facing; against; so that, in-order-to
	D36	rdi / anx	give, place; living, life	bestow, grant; enduring, persisting
	D60	wab	priest, pure one	administrator, the purified functionary
	G1	A	(phonetic complement)	(phonetic complement)
	G17	m	from, with, in	out of, releasing from; within
	G43	w	(phonetic complement)	(phonetic complement)
	H6	mAat	truth, justice, right order	truth-and-power, the binding force
	I9	f	his, him (suffix pronoun)	his, him (suffix pronoun)
	L2	bity	King of Lower Egypt	sovereign of the horizon
	M17	i	(phonetic complement)	(phonetic complement)
	M23	nsw	King of Upper Egypt	sovereign of the horizon
	N33	(det.)	grain, mineral (det.)	grain, mineral (det.)
	N35	n	of, to, for	of, to, for
	O1	pr	house, temple; stronghold	god-house; fastness, sealed-house
	R8	nTr	god, sacred	the divine one, the radiant; the luminous one
	S12	nbw	gold	divine-flesh
	S14	Hd	silver, money	brightness, white-metal
	X1	t	bread; (feminine ending)	bread; (feminine ending)
	Y1	dmD	total, sum	completeness, the gathered whole
	Z2	(plural strokes)	(plural marker)	(plural marker)
	Z4	(dual strokes)	(dual marker)	(dual marker)

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